

HDOC - Day 36

Basic String Operations Using Separate Functions

Algorithm

Repeat a menu. Count characters by scanning until the string terminator without strlen(); print each character in a loop; count alphabetic vowels and consonants separately; exit on option 4.

Test Case Table

1	First operation	Expected output obtained	Pass
2	Second operation	Expected output obtained	Pass
3	Third operation	Expected output obtained	Pass

C Program

```
#include <stdio.h>
void countChars(void);void printChars(void);void countVC(void);
int main(){int ch;do{printf("\n1.Count characters\n2.Print characters\n3.Vowels/consonants\n4.Exit\nChoice: ");scanf("%d",&ch);getchar();switch(ch){case 1:countChars();
void countChars(void){char a[200];int i=0;printf("Enter string: ");fgets(a,200,stdin);while(a[i]&&a[i]!='\n')i++;printf("Number of characters = %d\n",i);}
void printChars(void){char a[200];int i;printf("Enter string: ");fgets(a,200,stdin);for(i=0;a[i]&&a[i]!='\n';i++)printf("%c\n",a[i]);}
void countVC(void){char a[200],c;int i,v=0,k=0;printf("Enter string: ");fgets(a,200,stdin);for(i=0;a[i]&&a[i]!='\n';i++){c=a[i];if((c>='A'&&c<='Z')||(c>='a'&&c<='z')){
```

Compilation, Execution & Testing

```
puppie@Anvikas-MacBook-Air Desktop % clang day36.c -o day36
puppie@Anvikas-MacBook-Air Desktop % ./day36
Choice: 1 String: Hello World
Number of characters = 11
```

```
puppie@Anvikas-MacBook-Air Desktop % ./day36
Choice: 2 String: CSE
C
S
E
```

```
puppie@Anvikas-MacBook-Air Desktop % ./day36
Choice: 3 String: Hello World
Vowels = 3
Consonants = 7
```

Result: The program was compiled, executed and tested successfully.